# UNIVERSIDAD INTERCULTURAL PARA LA IGUALDAD

## Programación e Innovación Digital

## Materia: Base de Datos

## PROYECTO INTEGRADOR: ABRAZA UN ARBOL

**Integrantes:**

**Nohemi Ibarra Vazquez**

**Griselda Alejandra Macias Herrera**

**Patricia Lizet Avila Zarate**

**Docente:**

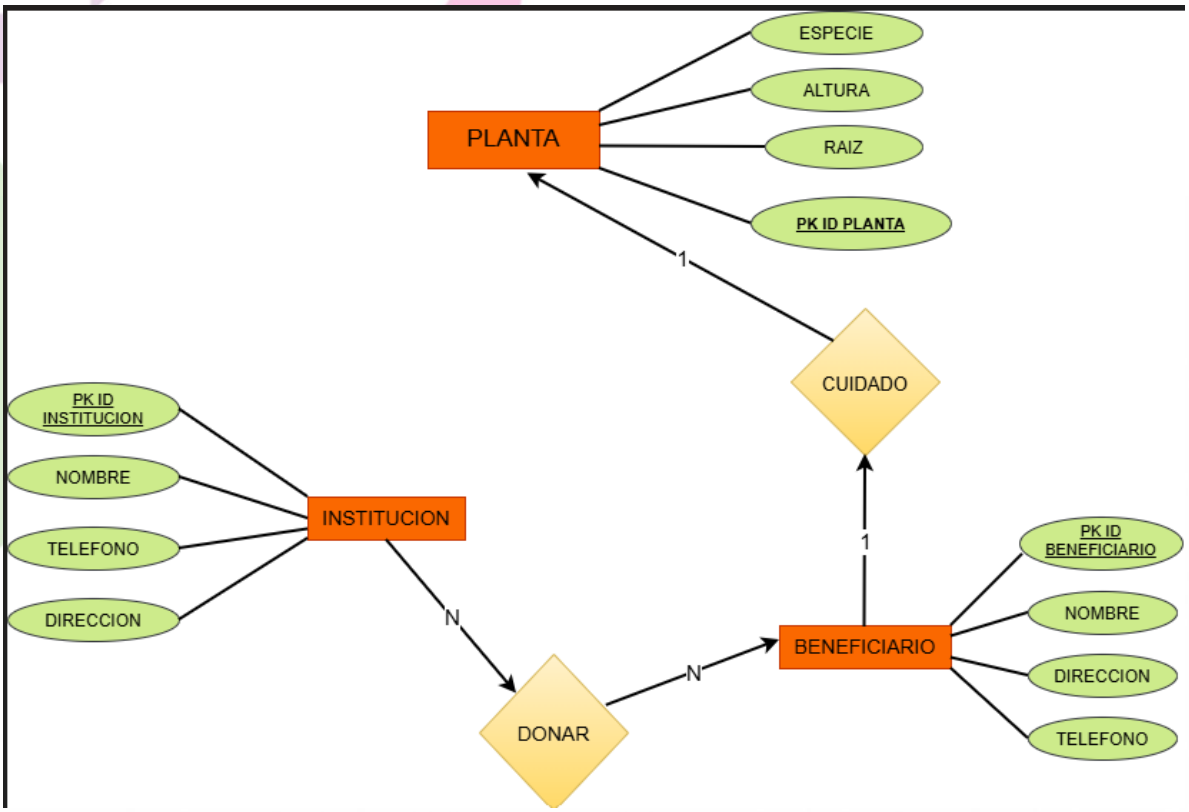
**Mtra. María del Carmen Castañeda**

**Fecha de entrega:**

**24 de octubre de 2026**

https://github.com/nohemiibarra1989z-lang/PROYECTO_ABRAZA_UN_ARBOL_DB

https://github.com/nohemiibarra1989z-lang/PROYECTO_ABRAZA_UN_ARBOL_DB.git




## TABLA PLANTA

| CAMPO | TIPO DE DATO | ROL | EXPLICACIÒN |
|---|---|---|---|
| ID_PLANTA | INT | PK | NUMERO UNICO PARA CADA PLANTA |
| ESPECIE | VARCHAR(100) | ATRIBUTO | NOMBRE PLANTA |
| ALTURA | VARCHAR(100) | ATRIBUTO | CUANTOS METROS CRECE |
| RAIZ | VARCHAR(100) | ATRIBUTO | TIPO DE RAIZ |

## TABLA BENEFICIARIO

| CAMPO | TIPO DE DATO | ROL | EXPLICACIÒN |
|---|---|---|---|
| ID_BENEFICIARIO | INT | PK | IDENTIFICACIÒN OFICIAL |
| NOMBRE | VARCHAR(50) | ATRIBUTO | IDENTIFICACIÒN |
| DIRECCIÒN | VARCHAR(100) | ATRIBUTO | UBICACIÒN FISICA |
| TELÈFONO | VARCHAR(15) | ATRIBUTO | CONTACTO |

## TABLA INSTITUCIÒN

| CAMPO | TIPO DE DATO | ROL | EXPLICACIÒN |
|---|---|---|---|
| ID_INSTITUCION | INT | PK | IDENTIFICACIÒN OFICIAL |
| NOMBRE | VARCHAR(50) | ATRIBUTO | IDENTIFICACIÒN |
| DIRECCIÒN | VARCHAR(100) | ATRIBUTO | UBICACIÒN |
| TELÈFONO | VARCHAR(15) | ATRIBUTO | CONTACTO |

## RELACIÒN 1:N

| CAMPO | TIPO DE DATO | ROL | EXPLICACIÒN |
|---|---|---|---|
| ID_INSTITUCION | INT | PK | IDENTIFICACIÒN OFICIAL |
| NOMBRE_INSTITUCION | VARCHAR(100) | ATRIBUTO | TEXTO CORTO |
| DIRECCIÒN | VARCHAR(100) | ATRIBUTO | UBICACIÒN |
| TELEFONO | VARCHAR(15) | ATRIBUTO | CONTACTO |
| ID_PLANTA | INT | FK | NUMERO UNICO PARA CADA PLANTA |

### TABLA PLANA INICIAL (SIN NORMALIZAR)

| ID_PLANTA | ESPECIE | ALTURA | RAIZ | ID_BENEFICIARIO | NOMBRE _BENEFICIARIO | DIRECCIÒN _BENEFICIARIO | TELÈFONO | ID_INSTITUCION | NOMBRE _INSTITUCIÒN | DIRECCIÒN_ INSTITUCIÒN | TELÈFONO |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 10 | TRUENO | 3 a 10metros | PIVOTANTE | 443 | ALEJANDRA HERRERA | GARAMBULLO #156 | 4492561305 | 22 | VIVERO MUNICIPAL | JARDINES DEL PARQUE | 4499723010 |
| 5 | TRUENO | 3 a 10metros | PIVOTANTE | 654 | MARISOL DE LEÒN | PITAHAYA #210 | 4494281620 | 46 | VIVERO LA PALMA | SOLIDARIDAD 1 | 4492739258 |
| 1 | TRUENO | 3 a 10metros | PIVOTANTE | 123 | ANDREA MACIAS | SAGUARO #135 | 4498034961 | 55 | VIVERO ESTATAL | ORTEGA DOUGLAS | 4491020976 |

### PRIMERA FORMA NORMAL (1FN)

| ID_PLANTA | ESPECIE | ALTURA | RAIZ | ID_BENEFICIARIO | NOMBRE _BENEFICIARIO | DIRECCIÒN _BENEFICIARIO | TELÈFONO |
|---|---|---|---|---|---|---|---|
| 10 | TRUENO | 3 a 10metros | PIVOTANTE | 443 | ALEJANDRA HERRERA | GARAMBULLO #156 | 4492561305 |
| 5 | TRUENO | 3 a 10metros | PIVOTANTE | 654 | MARISOL DE LEÒN | PITAHAYA #210 | 4494281620 |
| 1 | TRUENO | 3 a 10metros | PIVOTANTE | 123 | ANDREA MACIAS | SAGUARO #135 | 4498034961 |

## SEGUNDA FORMA NORMAL (2FN)

**PLANTA**

| ID_PLANTA (PK) | ESPECIE | ALTURA | RAIZ | ID_INSTITUCION (FK) |
|---|---|---|---|---|
| 10 | TRUENO | 3 a 10metros | PIVOTANTE | 22 |
| 5 | TRUENO | 3 a 10metros | PIVOTANTE | 46 |
| 1 | TRUENO | 3 a 10metros | PIVOTANTE | 55 |

**INSTITUCIÒN**

| ID_INSTITUCION (PK) | NOMBRE_INSTITUCIÒN |
|---|---|
| 22 | VIVERO MUNICIPAL |
| 46 | VIVERO LA PALMA |
| 55 | VIVERO ESTATAL |

**BENEFICIARIO**

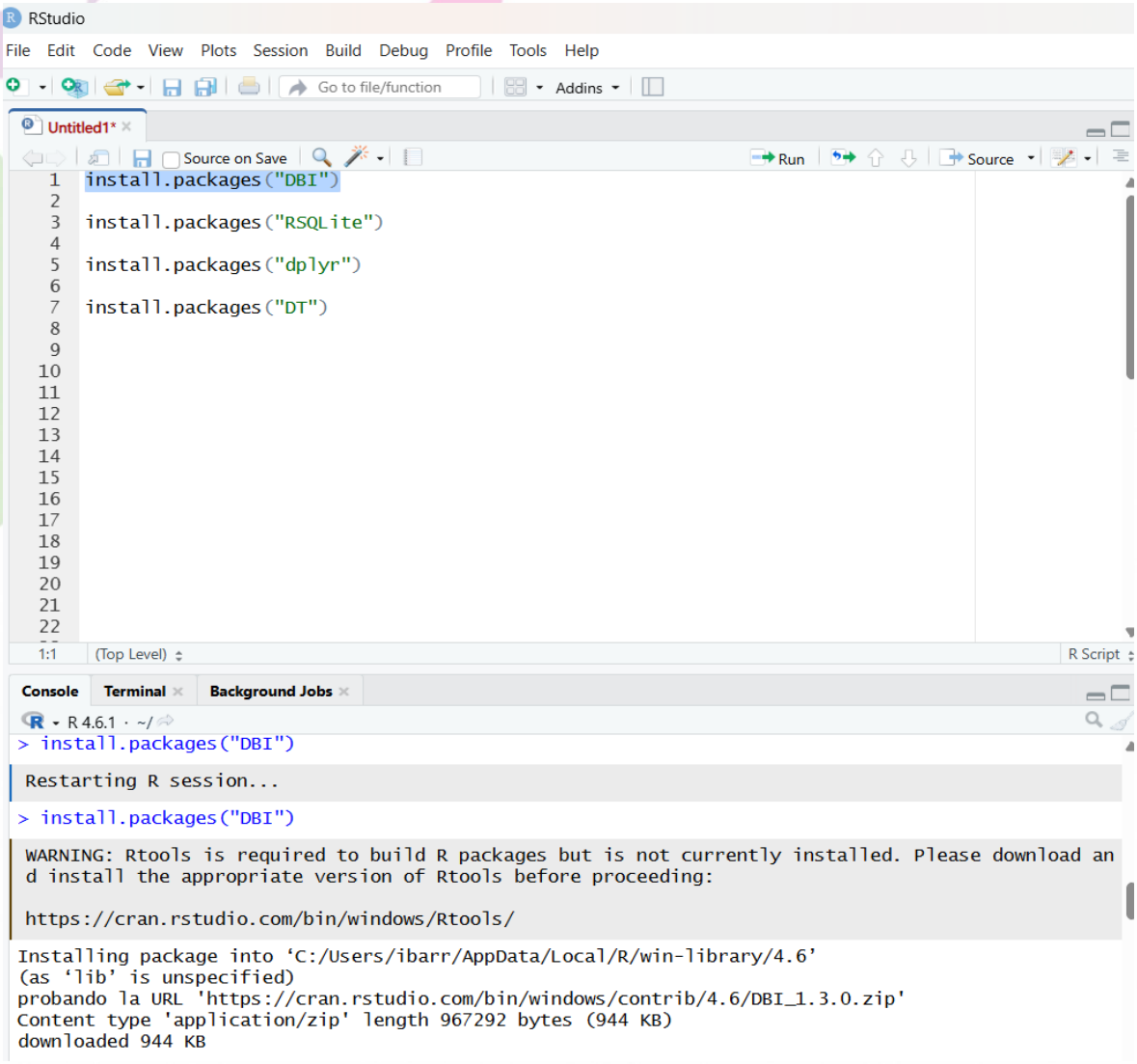
| ID_BENEFICIARIO (PK) | TELÈFONO_ CONTACTO | ID_PLANTA (FK) |
|---|---|---|
| 443 | 4492561305 | 10 |
| 654 | 4494281620 | 5 |
| 123 | 4498034961 | 1 |

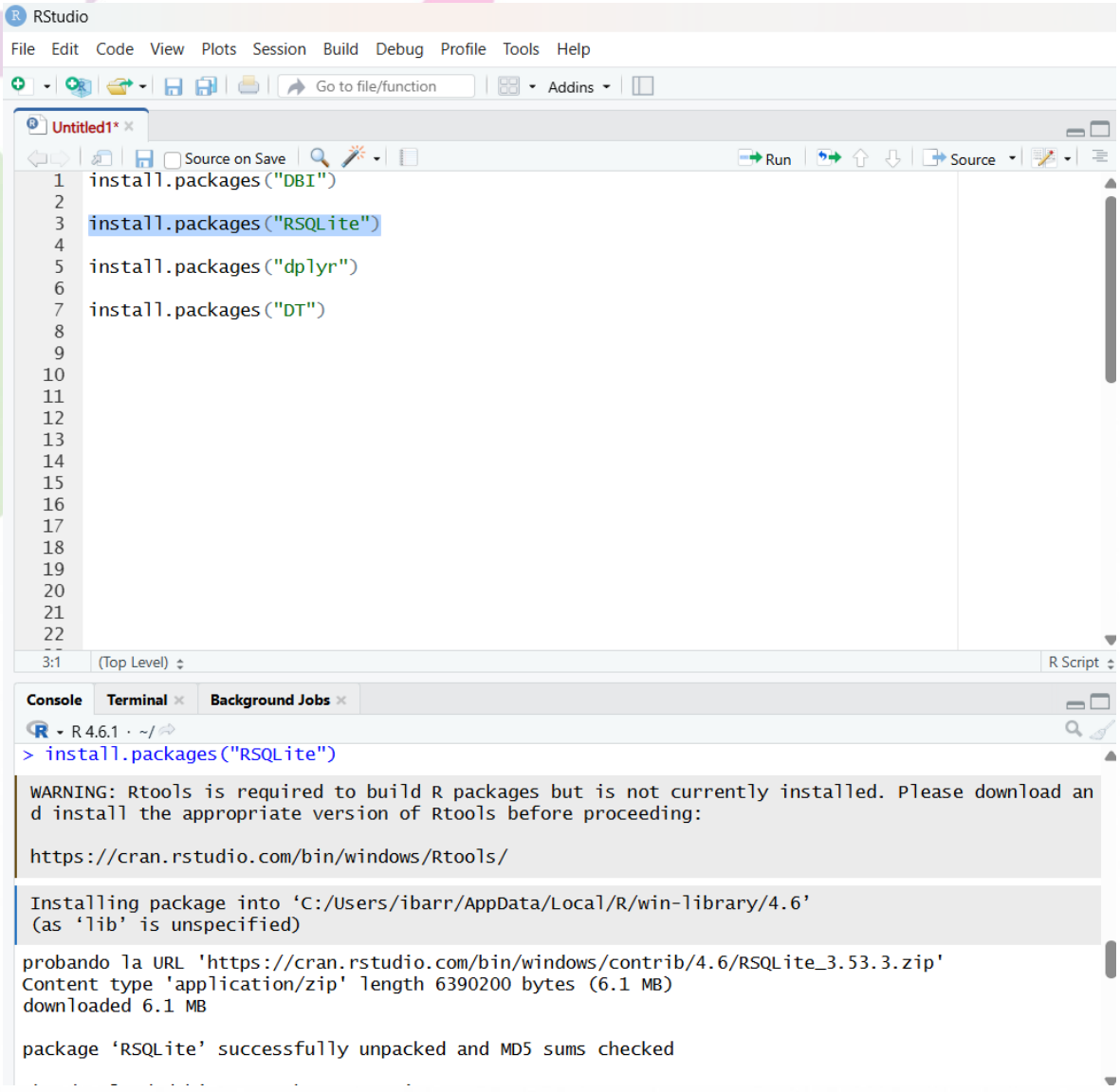
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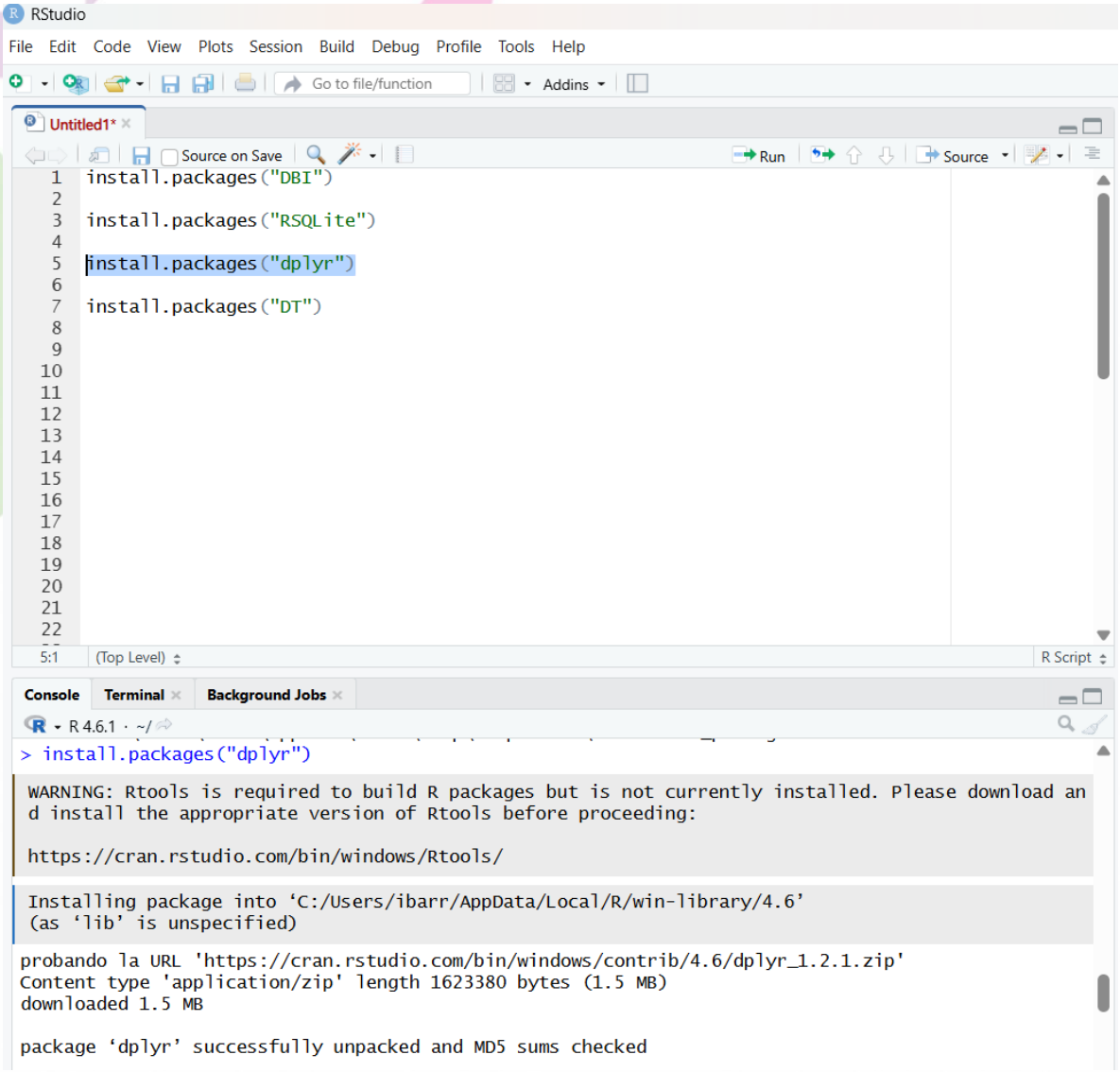
| PLANTA | | |
|---|---|---|
| PK | ID_PLANTA | INT |
| | ESPECIE | VARCHAR(100) |
| | ALTURA | VARCHAR(100) |
| | RAIZ | VARCHAR(100) |
| FK | ID_INSTITUCION | INT |

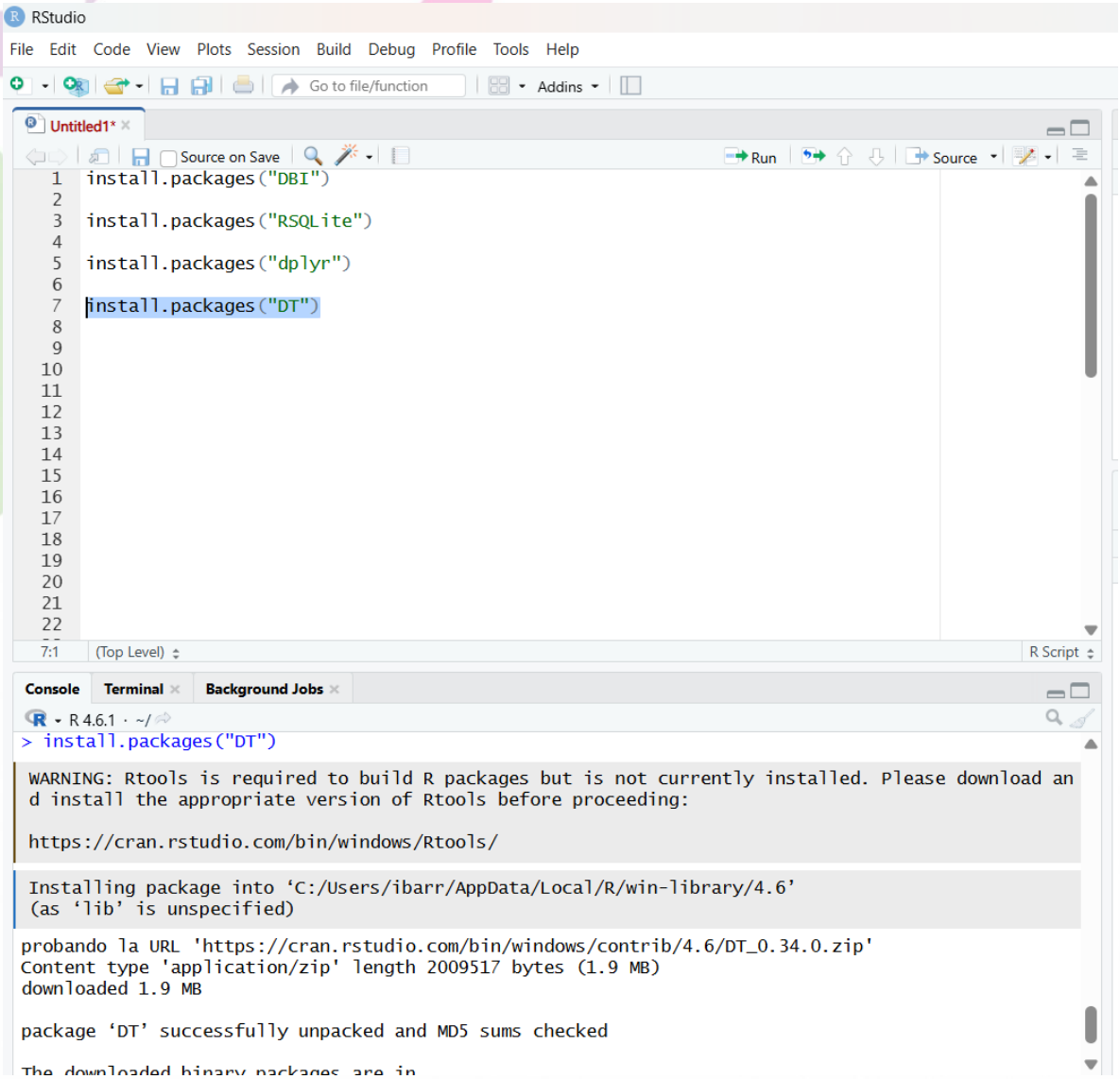
| BENEFICIARIO | | |
|---|---|---|
| PK | ID_BENEFICIARIO | INT |
| | NOMBRE | VARCHAR(50) |
| | DIRECCIÒN | VARCHAR(100) |
| | TELÈFONO | VARCHAR(15) |
| FK | ID_PLANTA | INT |

| INSTITUCIÒN | | |
|---|---|---|
| PK | ID_INSTITUCION | INT |
| | NOMBRE | VARCHAR(50) |
| | DIRECCIÒN | VARCHAR(100) |
| | TELÈFONO | VARCHAR(15) |
| FK | ID_BENEFICIARIO | INT |


RStudio
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Addins
Untitled1*
Source on Save
Run
Source
1 install.packages("DBI")
3 install.packages("RSQLite")
5 install.packages("dplyr")
7 install.packages("DT")
1:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> install.packages("DBI")
Restarting R session...
> install.packages("DBI")
WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:
https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/ibarr/AppData/Local/R/win-library/4.6'
(as 'lib' is unspecified)
probando la URL 'https://cran.rstudio.com/bin/windows/contrib/4.6/DBI_1.3.0.zip'
Content type 'application/zip' length 967292 bytes (944 KB)
downloaded 944 KB


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled1*
Source on Save
Run
Source
1 install.packages("DBI")
3 install.packages("RSQLite")
5 install.packages("dplyr")
7 install.packages("DT")
3:1 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> install.packages("RSQLite")
WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:
https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/ibarr/AppData/Local/R/win-library/4.6'
(as 'lib' is unspecified)
probando la URL 'https://cran.rstudio.com/bin/windows/contrib/4.6/RSQLite_3.53.3.zip'
Content type 'application/zip' length 6390200 bytes (6.1 MB)
downloaded 6.1 MB
package 'RSQLite' successfully unpacked and MD5 sums checked


RStudio
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Go to file/function
Addins
Untitled1*
Source on Save
Run
Source
1 install.packages("DBI")
3 install.packages("RSQLite")
5 install.packages("dplyr")
7 install.packages("DT")
5:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> install.packages("dplyr")
WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:
https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/ibarr/AppData/Local/R/win-library/4.6'
(as 'lib' is unspecified)
probando la URL 'https://cran.rstudio.com/bin/windows/contrib/4.6/dplyr_1.2.1.zip'
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downloaded 1.5 MB
package 'dplyr' successfully unpacked and MD5 sums checked


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled1*
Source on Save
Run
Source
1 install.packages("DBI")
3 install.packages("RSQLite")
5 install.packages("dplyr")
7 install.packages("DT")
7:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> install.packages("DT")
WARNING: Rtools is required to build R packages but is not currently installed. Please download an
d install the appropriate version of Rtools before proceeding:
https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/ibarr/AppData/Local/R/win-library/4.6'
(as 'lib' is unspecified)
probando la URL 'https://cran.rstudio.com/bin/windows/contrib/4.6/DT_0.34.0.zip'
Content type 'application/zip' length 2009517 bytes (1.9 MB)
downloaded 1.9 MB
package 'DT' successfully unpacked and MD5 sums checked
The downloaded binary packages are in